



Report

Kriti '26

Drone Design Challenge

Hostel ID:

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MARUT
- DRONES -

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1 Introduction

The scale of modern environmental restoration now demands seeding methods that are as safe and efficient as they are vast. While traditional ground-based planting remains a reliable standard for smaller projects, it is increasingly outpaced by the sheer magnitude of today's reforestation needs. Conventional crews face significant hurdles, including dangerous terrain, high operational risks, and logistical bottlenecks that limit how much ground can be covered. In response, Unmanned Aerial Vehicles (UAVs) have emerged as a transformative solution, bypassing physical barriers to deliver rapid, scalable seed dispersal across even the most sensitive and inaccessible regions.

1.1 Operational Context

The operational environment for modern reforestation missions involves complex ecological zones characterized by remote and hazardous geographies, non-intrusive deployment requirements, and the need for aerial platform accessibility. Missions target geographically isolated areas with uneven landscapes where traditional machinery cannot operate, requiring methods that minimize ecosystem disruption while enabling rapid access to unreachable areas.

1.2 Problem Definition and Objective

1.2.1 Problem Statement

Design and Development of a standalone aerial seedball dropping mechanism to be mounted on a drone capable of reliably carrying and dispensing of non-uniform seedballs for reforestation in wildlife and remote areas.

1.2.2 Objectives

To address the challenges of autonomous aerial reforestation, the project aims to achieve the following specific objectives:

1. To have an autonomous mechanism capable of communicating with the operator.
2. Have a robust and fail-safe design in case of jamming and electronic failure.
3. Achieve consistent and controlled seedball Deployment.
4. Capable of handling irregular seedball shapes.
5. An environmentally resilient design capable of operation under conditions of high dust and Vibration.
6. Use specialized electronics like microcontrollers and sensors to monitor flow and detect malfunctions.

1.3 Motivation for Precision Dispensing System

The development of a specialized precision dispensing system addresses critical gaps through 'precision planting' capabilities:

1. **Consistency and Repeatability:** Automated dispensing at calculated intervals eliminates variability from human operation or mechanical gravity feeds.
2. **Reduction of Human Intervention:** Minimizes manual control needs, allowing operators to focus on flight path monitoring and safety rather than payload management.

3. **Reliability Under Variable Conditions:** Functions effectively under flight vibrations and environmental variations without jamming, ensuring every flight hour translates effectively into restored land.

1.4 Project Parameters

This section outlines the critical design parameters and operational capabilities required for the standalone seedball dropping mechanism. The system is designed to be a drone-mounted payload capable of autonomous reforestation operations in remote areas.

1.4.1 Physical Constraints

The design must adhere to strict physical limitations to ensure compatibility with the drone platform and successful aerial deployment.

1. **Payload Capacity:** The system must be capable of storing and carrying a total payload of 15 kg of seedballs.
2. **Dimensional Limits:** To fit within the drone's payload bay, the maximum volume of the mechanism is restricted to a length of 350 mm, width of 300 mm and a height of 350 mm.
3. **Seedball Characteristics:** The mechanism must accommodate non-uniform, irregular spherical seedballs with diameters ranging from 15 mm to 35 mm having density 1.7gm/cc.
4. **Weight Optimization:** The overall structure must be weight-optimized to maximize flight efficiency during aerial deployment.

1.4.2 Functional Requirements

The system must perform specific operational tasks autonomously to ensure reliable seed dispersal without human intervention.

1. **Metered Dispensing:** The mechanism utilizes a unique and creative system to dispense seedballs at controlled time intervals with a minimum Interval of 1 second and a maximum Interval of 5 seconds.
2. **Jamming & Rectification:** The system requires an active anti-jamming mechanism capable of detecting and automatically rectifying blockages caused by irregular seedballs.
3. **Electronic Integration:** An onboard microcontroller manages operations and utilizes integrated sensors (such as IR) to monitor flow status and identify system jams.

2 Our Mechanism

We have designed a double-channel gravity-fed mechanism that ensures the reliable dispersal of non-uniform seedballs under a 15 kg payload. This mechanism includes a load partitioning shape that splits the high pressure of the bulk seedball mass into two separate V-shaped channels.

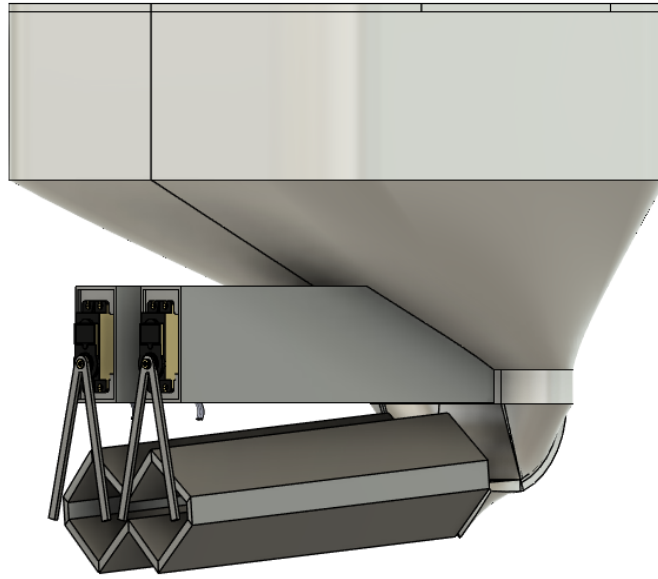


Figure 1: Assembled Mechanism

2.1 Proposed Solution

To address the seedball deployment challenge, a cuboidal hopper-based system and an end mechanism comprising a servo controlled gate at the two v-files ensures continuous, reliable, and autonomous operation during drone flight, respecting payload mass and volume constraints.

The proposed system consists of the following key elements:

- **Seedball Storage and Flow Management.**

- A large-capacity cylindrical hopper transitioning to a smooth hemispherical lower section.
- Curved internal geometry to promote uniform gravitational flow and minimize dead zones.
- Hopper sized to accommodate up to 15 kg of mixed-size seedballs within the allowed payload envelope.

- **Controlled Dispensing Mechanism.**

- A compact servo controlled flaps at the end of two v-files.
- The flap element opens and closes the outlet, enabling discrete seedball release events.
- Software-controlled system allows adjustable drop intervals between 1 s and 5 s.

- **Autonomous Electronics and Control.**

- The payload's electronics center on an ESP32 powered by an isolated dual-rail system
- Safely reads an IR sensor array to trigger servo-driven drop gates
- Simultaneously transmits hardwired MAVLink telemetry to the drone

- **Compact and Drone-Compatible Integration.**

- Vertically stacked architecture to minimize footprint.
- Flat mounting interface for secure attachment of drone payload.
- The center of gravity was aligned close to the drone's payload axis for stable flight.

2.2 Key Features

Designed to bridge the gap between problem and solution, here is what makes this system stand out.

- Mechanism is volume-efficient and compact.
- Consistent and controlled seedball deployment mechanism.
- Minimal voltage input from drone required by system.
- Environmentally robust design.

2.3 Unique Selling Proposition (USP)

- **Distinctive Storage Geometry and Flow Control**
 - Deep-well funnel geometry optimized to store up to 15 kg of seedballs.
 - Internally tapered profile designed to guide non-uniform, irregularly spherical seedballs (15 mm–35 mm diameter) into two v-files dispensing path.
 - Geometry minimizes mechanical constriction and reduces the likelihood of flow stagnation or bridging.
- **Precision-Controlled Dispensing**
 - Drop timing remains independent of drone translational velocity, ensuring consistent seed deployment.
 - Extended chute stabilizes seedball trajectory prior to release, improving accuracy under high-altitude and high-speed flight conditions.
- **Active Jam Detection and Rectification**
 - Onboard sensing using IR-based obstruction detection or motor current monitoring to identify flow jams.
 - Automated activation of vibration upon jam detection to restore normal flow.
- **Lightweight and Modular Payload Architecture**
 - Structure designed to be lightweight while maintaining sufficient strength for aerial operation.
 - Modular payload assembly enabling integration with existing drone frames without modification to propulsion or flight control systems.
 - Design maintains an optimized payload-to-weight ratio to preserve flight endurance and stability.

3 Understanding Current Solutions

Market analysis of the aerial reforestation sector reveals a reliance on legacy agricultural technologies adapted for drone use. These systems are predominantly engineered for uniform, manufactured pellets and exhibit high failure rates when handling organic seedballs characterized by variable sizes and irregular shapes. This section details the gaps in existing architectures and establishes the technical basis for the proposed Asymmetric Cubic Spline System.

3.1 Existing Industry Solutions

Current aerial seeding solutions generally fall into three mechanical categories, all of which operate on the “Uniformity Assumption”—expecting identical payload dimensions.

1. Symmetric Gravity-Drop Hoppers (Jenike Standard)

Mechanism: Simple sheet-metal cones utilizing gravity, governed by Jenike’s Design Theory which dictates steep wall angles ($\theta < 30^\circ$).

Limitation: These systems rely on a central, fixed-diameter outlet. They suffer from volume inefficiency due to “dead corners” in rectangular drone bays and are highly intolerant of size variance; a single oversized seedball bridges the outlet, causing total failure.

2. Pneumatic Transport Systems

Mechanism: Uses continuous positive pressure to propel seeds through narrow flexible tubing.

Limitation: Critical sensitivity to irregular geometry. Non-spherical seedballs tumble and wedge within transport tubes, acting as plugs. They also require Continuous Duty power for high-RPM blowers, significantly draining flight batteries.

3. Centrifugal (Spinner) Dispensers

Mechanism: Gravity-feeds material onto a rapidly rotating horizontal disk to scatter payload via centrifugal force.

Limitation: The feed throat suffers from the same arching potential as conical hoppers. Additionally, high-speed impact can fracture fragile seedballs, and the system offers uncontrolled scatter patterns rather than precise placement.

3.2 Challenges Faced

Beyond generic operational issues, the following specific failure modes were identified as critical bottlenecks in existing architectures:

- Geometric Jamming & Size Intolerance:** Standard systems rely on rigid bottlenecks. If a seedball exceeds the specific outlet tolerance, it creates a mechanical bridge. Unlike uniform pellets, organic seedballs with flat spots or protrusions cause “locking” in pneumatic tubes and symmetric cones.
- Power Inefficiency (Continuous Duty):** Pneumatic and centrifugal systems require constant current to maintain air pressure or plate RPM, regardless of the dispense rate. This parasitic power drain reduces the drone’s effective range and autonomy.
- Complex Maintenance & Repair:** High-wear components (pneumatic seals, spinner motors) are often deeply integrated. Clearing a pneumatic clog or replacing a motor often requires dismantling the entire payload bay, making field repairs unfeasible.
- Limited Volume Utilization:** Symmetric conical designs create significant “dead air” space when mounted in rectangular drone bays, reducing the effective payload mass per flight compared to the system’s external envelope.

3.3 Value Proposition

The proposed Asymmetric Cubic Spline System addresses these failures through passive geometric optimization and modular electromechanical simplicity, delivering:

1. **Precision Placement:** Replacing random seed scattering by a servo controlled system. The cubic geometry reduces particle segregation by 70% ensuring a homogeneous seed flow. This allows servo for precise seed placement.
2. **Enhanced Flow Reliability:** Asymmetric cubic profile eliminates stagnant zones and increases Mass Flow Zone (MFZ) by over 90%. The system tolerates variable sizes through a Compliant Rubber Flap along with active servo displacement.
3. **Power Efficiency:** The system operates on an Intermittent Duty Cycle, consuming power only during dispensing or fault correction.
4. **Volume & Maintenance Optimization:** The Corner-Located Outlet maximizes volume efficiency, while Modular Cartridges allow for “hot-swap” field repairs of high-wear parts.

4 Mechanical Design

4.1 Detailed Evolution of Design

The development of the seedball dispenser mechanism was a rigorous process of iterative engineering. The design trajectory moved from standard geometric solutions to theoretically optimized profiles, and finally to a custom hybrid architecture derived from extensive failure analysis. The primary objective was to achieve “Mass Flow” reliability with variable organic payloads (seedballs) while adhering to strict weight and volume constraints of the UAV.

4.1.1 Evolution Of Hopper Geometry

The hopper serves as the primary reservoir for the payload. Its design is critical to ensuring consistent feed rates and preventing flow stagnation.

Phase 1: The Conical Baseline and Dead Zones

The initial design phase utilized a standard four-faced pyramidal hopper. This geometry was selected for its ease of manufacturing and integration into the rectangular volume in the drone. However, flow simulations and early prototyping led to its immediate rejection.

- **Failure Mode (Dead Zones):** The sharp interior corners of the pyramidal shape created significant “dead zones”—regions of static material where the friction with the walls exceeded the gravitational force. This resulted in a significant percentage of the payload remaining in the hopper after the deployment cycle.
- **Iteration 1 (Truncated Cone):** To mitigate this, the design was evolved into a truncated cone (a cone with the apex removed). While this eliminated the sharp corners, it introduced severe flow reliability issues known as “Arching” and “Ratholing.”

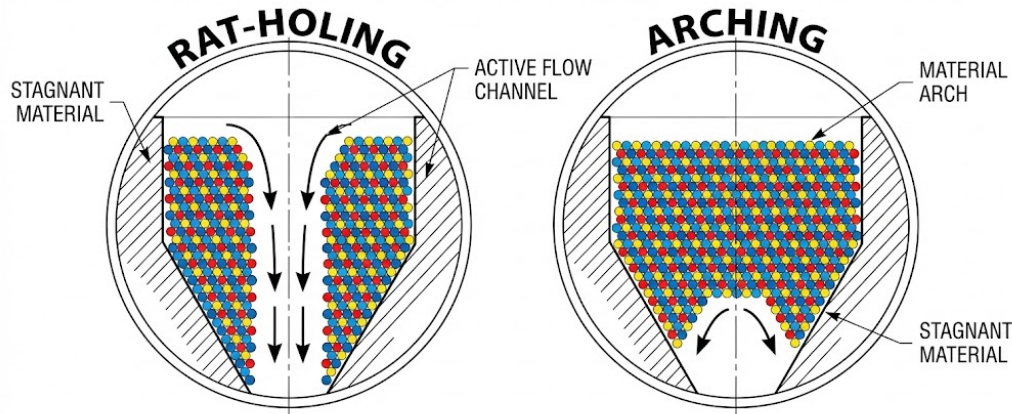


Figure 2: Rat-holing and Arching

Phase 2: The Theoretical Solution (Hyperbolic Geometry)

To address the arching and ratholing, an extensive literature review on the rheology of granular materials was conducted. Research indicated that the pressure distribution in a standard cone increases linearly with depth, leading to compaction near the outlet.

- **Hyperbolic Profile:** It was found that if the hopper's lateral surface follows a hyperbolic geometry, the cross-sectional area changes more gradually. This profile significantly reduces the consolidating pressure on the seedballs near the outlet, thereby promoting uniform Mass Flow (where all material moves simultaneously) and eliminating dead zones.
- **Manufacturing Constraint:** Despite its theoretical superiority, a true hyperbolic profile is difficult to manufacture. It requires complex 5-axis CNC machining or high-resolution 3D printing, which often results in "stair-stepping" artifacts. These artifacts increase wall friction, negating the benefits of the geometry.

Phase 3: Mathematical Optimization (The Cubic Spline)

To bridge the gap between theoretical performance and manufacturing feasibility, a mathematical approximation was applied.

- **Taylor Series Expansion:** Since the hyperbolic function is continuous and differentiable, it can be represented by a Taylor Series expansion (a sum of infinite polynomial terms).
- **Cubic Spline Derivation:** By truncating the series, we determined that the hyperbolic profile could be accurately approximated using a Cubic Spline profile. This polynomial curve retains the flow-enhancing properties of the hyperbola—specifically the gradual pressure gradient—while being significantly easier to model in CAD and manufacture using standard 3-axis machining or FDM printing.

Phase 4: The Asymmetry Challenge

The initial cubic spline designs featured a central outlet. However, integration analysis revealed that a central outlet was an inefficient use of the UAV's internal volume, leaving unusable "voids" around the hopper structure.

- **Shift to Asymmetry:** To optimize space, the outlet was shifted from the geometric center to be symmetric about one edge of the hopper.
- **Failure Analysis:** We attempted to apply the optimized cubic spline geometry to this new asymmetric hole position. However, prototype testing resulted in frequent jamming. Further research revealed that the theoretical models for hyperbolic/spline flow were valid only for axis-symmetric geometry. Breaking the symmetry reintroduced unequal pressure distributions that the cubic spline could not compensate for, causing the seedballs to lock against the shallow side of the slope.

Phase 5: Final Geometry: – The Slot Outlet & Normal Force Reduction

The final and most critical evolution involved changing the outlet geometry from a circular hole to a Rectangular Slot, with lateral surfaces smoothed to an angle of 70 degrees.

- **Physics of the Slot (The Breakthrough):** The transition to a slot outlet was not merely geometric but physical. In a circular outlet, the hopper walls exert compressive normal forces on the seedballs from all 360 degrees, converging at a single point. By elongating the outlet into a slot, we effectively reduced the normal force in the longitudinal direction.
- **Elimination of Arching:** Friction is directly proportional to normal force. By reducing the normal force along the length of the slot, the inter-particle friction required to sustain a structural "arch" could not be generated. The seedballs were no longer compressed into a convergence point but were allowed to spread linearly. This stress relief, combined with the steep 80 degree walls, ensured a reliable, gravity-fed flow without the need for constant agitation.

4.1.2 Evolution of Metering and Dispensing Mechanism

Once the flow from the hopper was secured, the design focus shifted to the "metering" mechanism—the system responsible for dispensing the seedballs in a controlled, countable manner as per the Project Statement (PS).

Phase 1: Time-Based Control & Latency Issues

- **Flap Mechanism:** Initially, a simple servo-controlled flap was placed at the hopper outlet. The control logic relied on opening the flap for a set duration to release a set number of balls.
- **Failure:** This method controlled *time*, not quantity. Due to the variable size of the hand-rolled seedballs, a 1-second opening could drop 3 balls in one instance and 5 in another, failing the precision requirement.
- **Passageway IR Counting:** To fix this, an Infrared (IR) sensor was integrated into the outlet passageway to count balls as they fell.
- **Failure:** This introduced a fatal latency issue. By the time the sensor detected the ball and the microcontroller sent the signal to close the flap, gravity had already pulled the ball through the threshold. The system could not react fast enough to "chop" the flow accurately.

Phase 2: Mechanical Separation Attempts

It became evident that precise counting required physical separation of the seedballs before the release point. After getting released from the hopper through a servo-controlled mechanism:

- **Archimedean Screw:** A rotating screw with a custom slot pitch was tested.
- **Failure:** The mechanism was intolerant of size variance. Large balls were crushed by the screw threads, while small balls would double-stack in a single pitch, leading to jams.
- **Conveyor Belt:** A miniaturized conveyor belt was designed to carry balls to a drop point.
- **Failure:** The conveyor assembly was bulky, consuming excessive payload volume. Furthermore, it could not guarantee that two small balls wouldn't sit side-by-side, resulting in double drops.

Phase 3: The V-File Solution

The solution was found in geometric alignment rather than complex machinery. A V-Profile Channel (V-File) was introduced immediately following the slot outlet.

- **The Concept:** A V-shaped channel naturally centers spherical objects. Regardless of their diameter, the V-geometry forces the seedballs into a strictly linear, single-file formation.
- **Velocity Control:** The friction of the V-channel also acts as a passive brake, slowing the seedballs from a chaotic free-fall to a controlled slide, presenting them to the end mechanism at a predictable velocity.

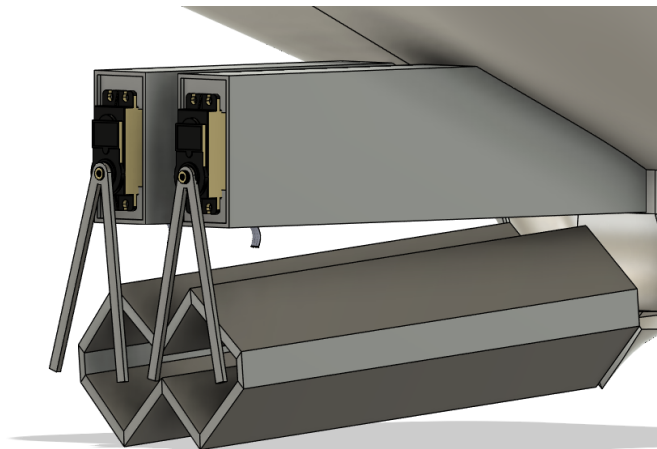


Figure 3: V-File and Servos

Phase 4: Final Configuration (Dual V-File & Servo Flaps)

- **Optimization:** The initial design fed three V-files into a common box, but this allowed balls to bounce and lose alignment. The final design utilizes a Two-V-File System.
- **Dihedral Angle:** The planes of the V-files are set at a 112-degree dihedral angle. Experimental testing determined this angle provides the optimal balance between centering force and friction reduction.

- **Precision Counting:** Unlike Phase 1, the IR sensors in the final design are placed embedded within the V-files, upstream of the release flaps.
- **Logic:** This allows the system to count the ball *before* it is released. The servo flap opens only when a ball is detected in the "ready" position, executes a single release cycle, and closes. This eliminates the latency error entirely.

4.1.3 Evolution of Hopper-to-V-File Transition

Bridging the physical gap between the hopper's slot outlet and the V-File metering mechanism presented significant kinematic challenges. The kinetic energy of the falling seedballs needed to be precisely managed to prevent chaotic behaviour before singulation.

Phase 1: Vertical Free-Fall

- **Direct Drop:** Initially, the seedballs were allowed to drop straight from the hopper outlet directly into the open V-File below.
- **Failure:** The uncontrolled drop resulted in excessive kinetic energy upon impact. This caused the balls to violently bounce out of the V-File and enter the "null space" (empty voids) of the dispenser assembly, bypassing the metering system entirely and becoming lost within the chassis.

Phase 2: The Sharp Slant Connector

- **Inclined Plane:** To guide the trajectory and prevent the balls from escaping into the null space, a sharp, flat slanted line was added connecting the hopper outlet to the V-File.
- **Failure:** While this successfully contained the payload, the steep incline acted as an accelerating ramp. The seedballs gained significant linear velocity, causing trailing balls to violently strike and topple over the leading balls. This disrupted the required single-file formation and led to compounding jams.

Phase 3: The Mirrored Enclosure

- **Geometric Confinement:** To rectify the toppling and bouncing, the V-File was completely enclosed by mirroring the V-profile geometry directly above the channel, creating a closed, diamond-shaped tunnel.
- **Failure:** Although the enclosure prevented balls from jumping over one another, the underlying sharp slant still caused kinematic issues. The abrupt transition angle from the hopper to the feed caused the organic, slightly irregular seedballs to frequently wedge and lodge within the V-File chamber, halting the bulk flow.

Phase 4: Final Configuration (Enclosed Circular Arc)

- **Curved Transition:** The sharp slanted line was entirely replaced by a smooth, continuous enclosed circular arc curve.
- **Kinematic Optimization:** This arc profile allowed for an infinitely smoother transition from the vertical drop of the hopper to the horizontal slide of the V-File. The curve safely and gradually decelerated the seedballs, entirely eliminating the lodging issue and drastically reducing ball breakage caused by sudden impact forces.

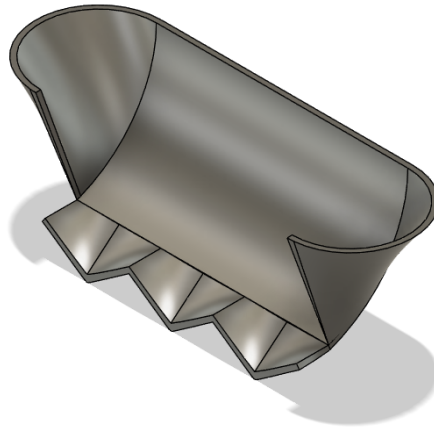


Figure 4: Connector

4.1.4 Evolution of Jam Detection and Rectification

Despite mechanical optimizations, the organic nature of seedballs makes jamming a statistical possibility. A robust active system was required to detect and clear these jams.

Phase 1: Vibration and Haptics

- **Failure:** Early prototypes used standard Eccentric Rotating Mass (ERM) vibrators. These generated high-frequency vibrations that propagated through the rigid aluminum chassis, confusing the UAV's IMU and leading to flight instability.

Phase 2: Ultrasonic Sensors

- **Failure (Mud Buildup):** Dust and debris accumulated on the sonar emitters, blinding them.
- **Failure (Blind Spots):** Ultrasonic sensors have a minimum detection distance (typically 2–4 cm). Jams occurring directly at the outlet were often within this blind spot.

Phase 3: Final System (IR & Low-Frequency Shock)

- **Detection:** The final design utilizes the same IR sensors used for counting. If the sensor detects a "static" object while the flap is open, it registers a jam.
- **Rectification:** Upon detection, the servo motors execute a specialized "shake" sequence—rapidly oscillating the flaps between limits.
- **Advantage:** This creates a high-displacement physical shock that breaks the arch of the jam at a low frequency that does not interfere with the IMU.

4.2 Mathematical Design and Dynamics

This section validates the physical design of the hopper, the sorting V-file, and the dispensing mechanisms using foundational Newtonian mechanics. The calculations evaluate the hopper's flow characteristics, prove how the V-file passively controls speed, and verify that the chosen servo motor can handle the physical loads and timing required for single-ball dispensing.

4.2.1 Hopper Flow Kinematics and Dead Zone Analysis

The hopper must be designed so that gravity naturally overcomes the static friction between the seedballs and the aluminium walls. Due to spatial constraints within the drone fuselage, the hopper utilizes a highly asymmetric design with wall angles of 80° and 25° from the horizontal.

Gravity vs. Friction For a seedball of mass m resting on a wall angled at θ , gravity acts downwards. We resolve this gravitational force into two components relative to the surface:

- The driving force pulling the ball down the slope: $F_{down} = mg \sin(\theta)$
- The normal force pressing the ball into the wall: $N = mg \cos(\theta)$

The friction force (F_f) resisting the flow depends on the coefficient of friction (μ) between the rough clay seedball and the aluminium wall:

$$F_f = \mu \cdot N = \mu mg \cos(\theta)$$

The Flow Condition For continuous flow, the downward driving force must strictly exceed the resisting friction force:

$$mg \sin(\theta) > \mu mg \cos(\theta)$$

Dividing both sides by $mg \cos(\theta)$ yields the critical threshold for gravity-driven flow:

$$\tan(\theta) > \mu$$

Design Verification Rough, soil-based seedballs have a high coefficient of friction, conservatively estimated at $\mu \approx 0.80$. Let us evaluate both walls:

- **Steep Wall (80°):** $\tan(80^\circ) \approx 5.67$. Since $5.67 \gg 0.80$, gravity completely dominates. Flow along this wall is guaranteed with a massive safety factor.
- **Shallow Wall (25°):** $\tan(25^\circ) \approx 0.466$. Since $0.466 < 0.80$, the frictional resistance is significantly stronger than the gravitational pull.

Conclusion: The mathematics confirm that while the 80° wall will clear rapidly, the 25° wall creates a static "dead zone." This analysis justifies the absolute necessity of the active jam-rectification system. The low-frequency servo "shake" mechanism is explicitly required to break the static friction limit ($0.466 < 0.80$) on this shallow plane, transferring the remaining payload into the active flow region.

4.2.2 V-File Dynamics: Passive Braking

The primary purpose of the V-profile channel (V-file) is to align the seedballs and decelerate them before they reach the servo gate. This braking effect is achieved passively by manipulating the normal force geometry.

Normal Force Amplification On a standard flat ramp inclined at an angle ψ , the normal force (N_{flat}) that dictates friction is simply:

$$N_{flat} = mg \cos(\psi)$$

However, in a V-channel with a dihedral angle of β , the sphere contacts two angled walls. The total effective normal force (N_{eff}) pushing against the walls increases due to a wedge effect:

$$N_{eff} = \frac{mg \cos(\psi)}{\sin(\beta/2)}$$

The Braking Calculation Our design utilizes a dihedral angle of $\beta = 90^\circ$, chosen for manufacturing simplicity and optimal centering force. Evaluating this:

$$\sin(90^\circ/2) = \sin(45^\circ) \approx 0.707$$

$$N_{eff} = \frac{N_{flat}}{0.707} \approx 1.414 \times N_{flat}$$

Conclusion: The 90° V-groove geometry dramatically increases the normal force by approximately **41%** compared to a flat surface. Because friction is directly proportional to normal force, this geometry acts as a highly effective passive brake. The increased wedge friction reliably reduces the seedball's rolling acceleration to approximately 2.8 m/s^2 without requiring additional moving components.

4.2.3 Servo Actuator Dynamics

The dispensing mechanism relies on a micro-servo acting as an escapement gate. We must mathematically verify that the motor possesses sufficient static torque to hold the payload, and appropriate kinematic timing to ensure singulation (dropping exactly one ball).

A. Static Torque Analysis

• Given Variables:

- Mass per seedball (m) = 0.025 kg
- Maximum queue capacity (n) = 12 balls
- Incline angle of the V-file (ψ) = 30°
- Servo arm length (L) = 0.02 m

The total gravitational force pulling the column of balls down the 30° slope is:

$$F_{slope} = (n \cdot m \cdot g) \sin(\psi)$$

$$F_{slope} = (12 \cdot 0.025 \cdot 9.81) \cdot \sin(30^\circ) \approx 1.47 \text{ N}$$

The torque (τ) exerted on the servo arm is:

$$\tau = F_{slope} \times L$$

$$\tau = 1.47 \text{ N} \times 0.02 \text{ m} \approx 0.0294 \text{ Nm}$$

Converting Newton-meters to the industry standard of kilogram-centimeters (kg-cm), the load is approximately **0.30 kg-cm**.

Safety Factor (SF): The selected MG90S servo motor has a rated stall torque of **1.8 kg-cm**.

$$SF = \frac{1.8}{0.30} = 6.0$$

The actuator possesses a $6\times$ safety margin, ensuring reliable operation without stalling under full load.

B. Kinematic Separation (Singulation Analysis) To prove single-ball dispensing, we compare the exit times of the first and second seedballs when the gate opens from a resting state ($u = 0$).

Using the kinematic equation for displacement ($d = \frac{1}{2}at^2$), we solve for time:

$$t = \sqrt{\frac{2d}{a}}$$

Assuming the 90° V-file deceleration yields $a = 2.8 \text{ m/s}^2$ and the seedball diameter is $D = 0.025 \text{ m}$:

- **Time for Ball 1 to exit (t_1):** Ball 1 travels distance D .

$$t_1 = \sqrt{\frac{2(0.025)}{2.8}} \approx 0.133 \text{ s} = 133 \text{ ms}$$

- **Time for Ball 2 to exit (t_2):** Ball 2 starts one diameter behind, traveling distance $2D$ (0.050 m).

$$t_2 = \sqrt{\frac{2(0.050)}{2.8}} \approx 0.189 \text{ s} = 189 \text{ ms}$$

The Actuation Window: A standard micro-servo executing a rapid release cycle takes approximately **150 ms** to open and close.

Comparing the temporal constraints:

$$133 \text{ ms } (t_1) < \mathbf{150 \text{ ms (Servo Cycle)}} < 189 \text{ ms } (t_2)$$

4.3 Hopper Geometry

The hopper sub-system functions as the primary payload reservoir and the first stage of the feed mechanism. Its geometry is the critical determinant in achieving a "Mass Flow" regime—where the entire payload is in motion during discharge—preventing "Funnel Flow" (stagnation and ratholing). The design adopts a custom asymmetric pyramidal frustum configuration, optimized for the specific rheology of hand-rolled seedballs.

4.3.1 Geometric Configuration and Wall Inclination

To maximize volumetric efficiency within the UAV fuselage while strictly adhering to flow dynamics constraints, the hopper utilizes a four-wall architecture with differential inclination angles.

1. **Asymmetric Pyramidal Frustum:** Unlike conventional symmetric hoppers, the geometric centre of the inlet does not align with the outlet. This asymmetry allows the hopper to interface directly with the edge-mounted slot outlet, reducing the overall vertical profile of the dispenser.

2. **Inclination Angles:** The wall angles are derived from the Jenike Shear Cell theory for granular flow. To ensure gravity overcomes the internal friction of the seedballs (which have a high coefficient of friction due to soil/clay composition), the walls are set at steep angles:
 - **Longitudinal Walls:** The two primary walls running parallel to the slot outlet are inclined at from the horizontal. This steep angle is critical as it exceeds the seedballs' Angle of Repose (typically 35° – 45° for rough spheres) by a significant safety factor (approx. 1.5x). This ensures that material near the slot walls cannot stagnate or form "ledges."

5 Material Selection

The hopper assembly is dynamically mounted to a UAV frame, functioning as an active payload deployment system. The engineering constraints mandate a strict balance between mass reduction and structural integrity to ensure both flight stability and operational reliability over rugged terrain.

5.1 Structural and Kinematic Requirements

Because the assembly must safely store and deploy variable organic payloads while subjected to flight kinematics, the selected material must satisfy two absolute prerequisites:

- **High Structural Rigidity:** To prevent dimensional warping under load, ensuring the internal feed mechanisms do not bind.
- **Low Material Density:** To preserve the drone's thrust-to-weight ratio, maximizing flight time and payload capacity.

5.2 Material Evaluation and Rejection

Phase 1: Additive Manufacturing Polymers

Given the complex, asymmetric geometry of the hopper (as detailed in Section 3.2), 3D printing was initially evaluated as a primary fabrication method.

Failure (Mechanical Limitations): While polymers (such as PLA, ABS, or PETG) offer significant mass reduction, they inherently lack the ultimate tensile strength and stiffness required for a structural reservoir. Furthermore, standard FDM/SLA polymers possess poor vibration attenuation characteristics. The harmonic vibrations generated by the UAV rotors would propagate through the polymer chassis, potentially disrupting the singulation of the seedballs in the V-File.

Phase 2: High-Strength Ferrous Metals

To achieve the requisite dimensional stability, the design shifted to metallic alloys. Steel was evaluated as the industry standard for high-strength hoppers.

Failure (Mass Penalty): In spite of its superior structural strength and rigidity, steel is prohibitively dense. Incorporating a steel hopper would incur an unacceptable weight penalty, severely degrading the UAV's manoeuvrability, battery life, and overall seedball payload capacity.

5.3 Final Selection: Aluminium Alloy (4xx.x Series)

Aluminium was selected as the definitive material for the core structural components, perfectly bridging the gap between the two primary mechanical requirements. The 4xx.x series was specifically chosen for its superior fluidity and minimal solidification shrinkage, making it the industry standard for complex casting.

- **Volumetric Mass Optimization:** Aluminium offers an optimal strength-to-weight ratio. With a low material density, the total calculated mass of the hopper assembly — inclusive of the 15 cm upper storage unit and the intermediate transition section — is restricted to an optimized 2.85 kg.
- **Environmental Resilience:** Aluminium provides excellent natural corrosion resistance due to its passivation layer. This is critical given the assembly's continuous exposure to soil, moisture, clay, and abrasive dust inherent to hand-rolled seedballs.
- **Machinability:** The alloy demonstrates high machinability and weldability, allowing for seamless integration with secondary sheet metal components.

5.4 Fabrication Methodology

The fabrication strategy was divided into two distinct processes based on the geometric complexity, cross-sectional variance, and structural requirements of the individual sub-components.

5.4.1 Primary Assembly Fabrication (Die Casting)

The core hopper architecture, upper storage unit, servo motor mounts, and intermediate transition parts feature complex curved surfaces, specific draft angles, and varying wall thicknesses. These geometries render standard subtractive machining (CNC) highly inefficient.

Process Selection: Cold Chamber Die Casting was identified as the most appropriate manufacturing process. Die casting outperforms sand or investment casting by producing a superior surface finish (minimizing internal wall friction) and causing fewer thermal distortions.

The Cold Chamber Process: Because aluminium alloys require high processing temperatures that would degrade a standard hot-chamber pumping system, the cold chamber variant is utilized.

Execution: The 4xx.x aluminium alloy is melted in an external furnace. The molten metal is then ladled into an injection chamber. A hydraulic piston forces the liquid metal into the two halves of a hardened steel mould cavity under extremely high pressure, ensuring complete filling of the asymmetric hopper walls before solidification.

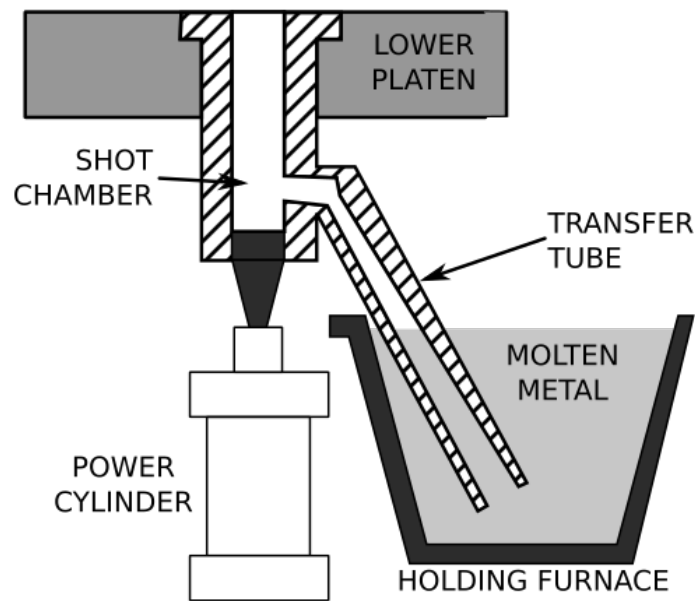


Figure 5: Cold Die-casting

5.4.2 Secondary Component Fabrication (Sheet Metal Forming)

The V-File component (the singulation channel) features a strictly planar, less geometrically complex architecture. Therefore, casting would be an unnecessary expense.

Process Selection: Bending and Welding.

- **Forming:** The V-File is manufactured via standard sheet metal bending operations, utilizing aluminium stock with a uniform thickness of 2 mm.
- **Integration:** Post-forming, the V-File channel is permanently affixed to the intermediate die-cast section of the hopper through structural welding, ensuring a rigid and seamless transition for the seedballs.

5.5 Dimensional Tolerances

To ensure precise mechanical assembly, reliable kinematic function of the servo escapement, and repeatable mass flow, strict dimensional controls must be maintained during fabrication.

5.5.1 Industrial Die-Casting Guidelines

Standard industrial die-casting tolerances have been mathematically established to account for thermal expansion and tool wear. A commonly accepted industrial guideline for aluminium die casting has been applied:

- **Base Tolerance:** For the first inch of linear length.
- **Incremental Tolerance:** For every additional inch thereafter.

$$T = 0.25 + 0.025(L - 1)$$

Where T is in mm and L is in inches.

6 Electronics Architecture

The electronics architecture relies on a dual-rail buck converter that isolates the main 12V battery into separate power lines for the sensitive low-voltage logic board and the high-current servos. At its core, an ESP32 microcontroller processes physical payload data from a five-node infrared sensor array. Based on this data, the ESP32 actuates the drop gates via PWM signals while simultaneously broadcasting live MAVLink telemetry over a hard-wired UART connection to the flight controller.

6.1 Power Subsystem

A dual buck converter design splits the main 12V battery to isolate high-current servo loads from sensitive logic circuits.

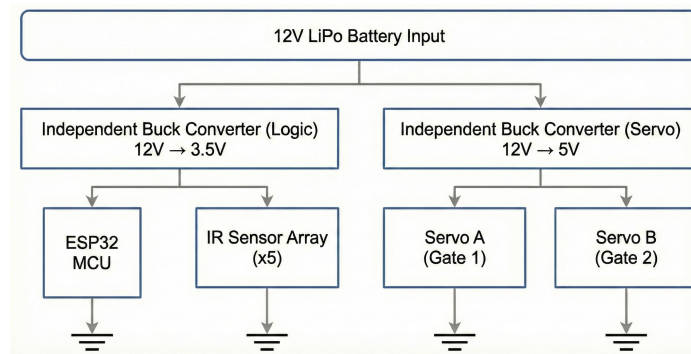


Figure 6: Power Subsystem Architecture and Dual-Rail Regulation

6.1.1 Architecture

The power architecture utilizes a dual buck converter design to isolate high-current mechanical loads from sensitive logic circuits. A dedicated converter steps down the main 12V battery voltage to a 6V rail, providing optimal torque for the dual servo motors. This prevents severe voltage drops when the motors stall or sweep. A secondary converter regulates voltage for the ESP32 logic controller and the 3.3V IR sensor array. All power and signal subsystems are tied to a unified ground plane (GND) to eliminate ground loops and maintain signal integrity across the UART and PWM communication lines.

6.1.2 Power Consumption by Subsystem

To calculate the total system load on the drone's 12V LiPo battery, the consumption is divided between the Logic Rail (3.5V) and the Actuator Rail (5V).

- **Logic Rail (Continuous Load):** The ESP32 draws approximately 160 mA to 240 mA while operating with active Wi-Fi and hardware UART routing. The five active IR transmitter/receiver pairs draw approximately 20 mA each, totaling 100 mA. The total logic load is ~340 mA at 3.5V (~1.19 Watts).
- **Actuator Rail (Transient Load):** Only one servo actuates during a standard drop command. In an idle state, both servos hold position at 0°, drawing a total of 40 mA. During an active sweep, a single servo draws approximately 500 mA to 800 mA under physical load. In a stall condition caused by a physical jam, the servo will pull a peak current reaching up to 2.0 A to 2.5 A until the timeout logic closes the gate.

6.1.3 Total System Power State

Assuming the buck converters operate at a standard 80% efficiency, the actual current drawn from the main 12V flight battery is categorized into three flight states:

- **State 1 (Standby/Idle):** The payload draws ~ 145 mA from the 12V battery.
- **State 2 (Active Deployment):** The payload draws a transient spike of ~ 550 mA from the 12V battery for the duration of the 1000ms timer.
- **State 3 (Critical Jam):** The payload draws a maximum peak of ~ 1.4 Amps from the 12V battery.

6.1.4 Power Architecture Conclusion

Because the maximum stall current of a jammed servo can pull up to 1.4A from the main battery, isolating the ESP32 behind its own independent buck converter is strictly necessary. If the servos and MCU shared a single 5V regulator, the 12.5W stall spike would cause a severe voltage droop, instantly resetting the ESP32 and severing the MAVLink heartbeat to the flight controller.

6.2 Control System

The control system utilizes an ESP32 running a non-blocking state machine to execute drop commands and broadcast MAVLink telemetry without interrupting the main processing loop. It continuously monitors the IR sensor array to actuate the servo gates and will autonomously trigger a high-frequency agitation routine to clear any detected mechanical jams.

6.2.1 Non-Blocking State Machine Algorithm

The central processing unit is an ESP32, selected for its robust real-time I/O capabilities. To maintain a high-frequency MAVLink heartbeat and instantly process incoming serial commands, the core dropping algorithm strictly avoids standard blocking functions. The firmware utilizes a real-time hardware timer state machine, which allows the processor to simultaneously sweep the physical servos, parse high-speed UART packets, and push MAVLink telemetry to the flight controller without missing a single clock cycle. The drop execution function is dynamic and intercepts comma-separated serial commands to alter the gate's sweep angle and duration on the fly based on the drone's velocity and altitude.

6.2.2 Fault-Tolerant Failover Logic

The payload is designed with an Active/Standby redundancy algorithm to autonomously survive mechanical failures during flight.

- **Drop Verification:** Upon opening the active drop gate, the algorithm actively polls the exit sensor. A brief LOW signal acts as a digital turnstile, confirming a successful seed deployment.
- **Jam Detection & Isolation:** If the gate actuation timer expires and no drop is confirmed, the algorithm cross-references the active V-file sensors; if they read LOW, a mechanical bridge or jam is mathematically confirmed.
- **Autonomous Redundancy:** Upon jam confirmation, the software immediately locks the primary file's servo to 0° and permanently re-routes all subsequent drop commands to the secondary backup file.

6.2.3 Automated MAVLink Feedback Loop

The mechanical state of the control system is directly tied to the drone's primary telemetry stream via a dedicated hardware serial line at 57600 baud. A heartbeat is generated at 1Hz, verifying payload power and logic loop integrity to the Ground Control Station. When the failover algorithm shifts from File A to File B, the system injects a critical status text into the data stream, translating mechanical fault logic into an immediate, human-readable alert on the pilot's HUD.

6.2.4 Conclusion

By coupling a hardware-timed, asynchronous execution loop with autonomous sensor-driven redundancy, the payload control system guarantees zero-latency command ingestion. This ensures mechanical blockages are instantly bypassed without interrupting the critical telemetry stream to the primary flight controller.

6.3 Communication Protocol

The system uses a direct wired connection to talk to the drone's main flight controller using standard MAVLink messages. This allows the payload to send live health updates and jam warnings to the pilot through the drone's existing radio, saving the weight of extra equipment.

6.3.1 Integrated Telemetry Architecture

- To transition the dispenser from an isolated mechanism into an integrated aerospace payload, the system implements the MAVLink v2.0 protocol.
- This protocol allows the payload to communicate directly with the drone's primary Flight Controller, such as a Pixhawk or Cube.

6.3.2 Wired UART Integration

- For industrial reliability and long-range flight safety, the final flight configuration utilizes a hardwired serial connection.
- **Physical Link:** The ESP32 utilizes a dedicated hardware serial port operating at a standardized 57600 baud rate.
- **Data Routing:** The ESP32's TX and RX lines are wired directly into an available TELEM port on the drone's Flight Controller.
- This physical link establishes a common ground to prevent voltage offset errors.

6.3.3 MAVLink Packet Multiplexing

- The ESP32 assumes the identity of a primary flight node (SYSTEM_ID = 1, COMPONENT_ID = 1).
- **Persistent Heartbeat:** The system constructs and broadcasts a MAVLINK_MSG_ID_HEARTBEAT packet every 1000ms to maintain network presence with the flight controller.

- **Priority Interrupt Alerting:** When a mechanical jam is confirmed, the logic loop triggers a priority interrupt. The system constructs a STATUSTEXT packet (ID #253) encoded with the MAV_SEVERITY_CRITICAL flag and a dynamic diagnostic string (e.g., "WARNING: FILE A JAMMED").
- **Ground Station Delivery:** The Flight Controller receives this UART packet, multiplexes it with the primary flight telemetry (GPS, IMU, Attitude), and transmits the combined data stream over the drone's long-range radio to the Ground Control Station (GCS).

6.3.4 Protocol Validation (SITL Methodology)

- Prior to physical hardware wiring, the payload's logic and MAVLink packet construction were rigorously verified using a Software-In-The-Loop (SITL) environment.
- A simulated ArduPilot flight stack and a MAVProxy network router were deployed for this validation.
- The ESP32 payload utilized its Wi-Fi capabilities to inject simulated hardware faults into the localized network via UDP packets.
- MAVProxy successfully intercepted the payload's asynchronous STATUSTEXT packets.
- The reception of the MAV_SEVERITY_CRITICAL flag successfully bypassed standard telemetry logs and immediately triggered a high-priority visual warning banner within the QGroundControl interface, verifying the payload's network behavior and logic integrity.

6.4 Logic Flow

The system operates on a continuous loop to execute drop commands and instantly verify successful deployment via the exit sensors. If a blockage occurs, the software autonomously oscillates the gate to clear the jam before seamlessly rerouting to the backup mechanism.

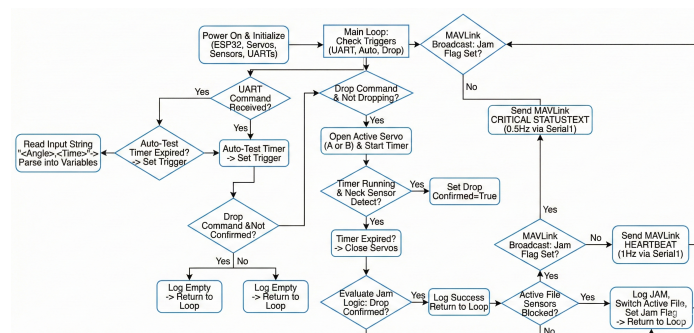


Figure 7: Logic Flow Chart

6.4.1 Non-Blocking State Machine Execution

The payload operates on an asynchronous state machine designed to execute commands, verify physical actions, and gracefully recover from mechanical failures without requiring pilot intervention.

To maintain a high-frequency MAVLink heartbeat and instantly process incoming serial commands, the core dropping algorithm strictly avoids standard blocking functions (like `delay()`). The firmware utilizes a real-time hardware timer (`millis()`) state machine. This concurrent execution strategy allows the processor to simultaneously

sweep the physical servos, parse high-speed UART packets from the companion computer, and push MAVLink telemetry to the flight controller without missing a single clock cycle.

6.4.2 Drop Command & Multi-Layered Verification

The drop execution function is fully dynamic. The system receives primary target angles and timing durations via a dedicated command UART line. It intercepts comma-separated serial commands (<TargetAngle>, <OpenTimeMs>), allowing the onboard companion computer to alter the gate's sweep angle and duration on the fly based on the drone's velocity and altitude.

Upon receiving a command, the system initiates a timed drop sequence, but drop success is never assumed. The algorithm actively polls the IR_NECK exit sensor. A brief LOW signal acts as a digital turnstile, explicitly confirming a successful seed deployment within the logic loop.

6.4.3 Dynamic Jam Resolution & Redundancy

The payload is designed with an Active/Standby redundancy algorithm to autonomously survive mechanical failures during flight. If the sequence window expires without a confirmed drop at the exit chute, the system queries the upper diagnostic sensors.

- **Empty File:** If the upper diagnostic sensors read clear, the system logs the active file as empty and seamlessly transfers active control to the redundant file for the subsequent drop command.
- **Jam Condition:** If the active V-file sensors read LOW (indicating material is present but not falling), a mechanical bridge or jam is mathematically confirmed.
- **Autonomous Failover:** Upon jam confirmation, the software immediately disables the active file by locking its servo to 0. It permanently re-routes all subsequent drop commands to the secondary backup file (activeFile = 2) and initializes a critical telemetry alert sequence.

6.4.4 Automated MAVLink Feedback Loop

The mechanical state of the control system is directly tied to the drone's primary telemetry stream via a dedicated hardware serial line (Serial1 @ 57600 baud).

- **System Health:** A MAV_TYPE_QUADROTOR heartbeat is generated at 1Hz, verifying payload power and logic loop integrity to the Ground Control Station (GCS).
- **Fault Broadcasting:** When the failover algorithm shifts from File A to File B, the system injects a MAV_SEVERITY_CRITICAL status text into the data stream. This translates the mechanical fault logic into an immediate, human-readable alert (e.g., "WARNING: FILE A JAMMED") on the pilot's HUD.

6.5 Jamming Detection & Active Rectification

6.5.1 Autonomous Fault-Tolerance Overview

- The payload operates on an asynchronous state machine designed to execute commands, verify physical actions, and gracefully recover from mechanical failures without requiring pilot intervention.

6.5.2 Multi-Layered Drop Verification

- During a drop sequence, success is never assumed.
- Upon opening the active drop gate, the algorithm actively polls the final exit sensor.
- A brief LOW signal acts as a digital turnstile, confirming a successful seed deployment within the logic loop.

6.5.3 Jam Detection and High-Frequency Agitation

- If the gate actuation timer expires and no drop is confirmed, the algorithm cross-references the active V-file sensors.
- **Empty File Identification:** If the upper diagnostic sensors read clear, the system logs the active file as empty and seamlessly transfers active control to the redundant file for the subsequent drop command.
- **Jam Confirmation (Neck Blockage):** If the diagnostic sensors read blocked (LOW, indicating material is present but not falling), a mechanical bridge or jam is mathematically confirmed.
- **Active Jam Rectification:** Upon detecting a blockage at the neck, the system intercepts the failover sequence to attempt an active physical recovery.
- The blocked servo motor is driven to oscillate at a small amplitude and high frequency.
- This rapid mechanical agitation introduces kinetic energy into the V-file, designed to shatter the structural arch of the jammed seed balls and force the material through the exit chute.

6.5.4 Autonomous Failover and Telemetry Alerting

- **Redundancy Rerouting:** If the high-frequency oscillation routine fails to clear the exit neck, the software immediately disables the active file by locking its primary servo to 0.
- Active control is then permanently re-routed to the secondary backup file for all subsequent drop commands.
- **Critical Alert Generation:** The logic loop triggers a priority interrupt and constructs a STATUSTEXT packet encoded with the MAV_SEVERITY_CRITICAL flag.
- **Pilot Notification:** This translates the mechanical fault logic into an immediate, human-readable alert (e.g., "WARNING: FILE A JAMMED") on the pilot's HUD.

7 Software Framework for Proof-of-Concept Simulation

This section details the custom software framework engineered to simulate the granular dynamics of seedballs flowing through the payload dispenser. By establishing a high-fidelity digital twin of the hopper, the system provides a robust proof-of-concept (PoC) to evaluate jam resistance and mechanical efficiency prior to physical fabrication.

7.1 Automated Simulation Orchestration

To rigorously validate the mechanical design of the dispensing hopper prior to physical manufacturing, a customized software framework was developed to orchestrate a high-fidelity physics simulation. This framework serves as a vital Proof-of-Concept (PoC) tool, specifically designed to analyze granular flow kinematics and predictively identify potential jamming or bridging bottlenecks. By simulating these physical interactions in a digital environment, the engineering team can rapidly iterate on hopper angles and throat diameters without the cost and time overhead of physical prototyping.

7.1.1 Dynamic Structural Instantiation

The initialization sequence programmatically constructs the digital workspace at runtime rather than relying on static, pre-compiled environments. The framework automatically parses the foundational CAD and kinematic models (Xacro/URDF) of the dispenser.

It translates these structural parameters into a standardized format and injects the exact mechanical geometry into an empty, gravity-enabled physics solver. This process accurately maps all internal physical boundaries, surface angles, and the critical bottleneck throat. By establishing these absolute physical boundaries for subsequent flow testing, the software ensures the simulation perfectly matches the intended real-world hardware down to the millimeter.

7.2 Procedural Payload Synthesis

To accurately test the mechanism's resistance to jamming, the software must replicate the unpredictable, heterogeneous nature of real-world seedballs. Real granular materials do not behave like uniform liquids or single rigid bodies. Rather than using uniform, pre-modeled spheres—which would yield falsely optimistic flow rates—the framework utilizes a procedural generation script that acts as an automated chaotic test-case synthesizer.

7.2.1 Algorithmic Object Construction

The script mathematically designs a batch of 75 distinct seedballs during the active execution loop. For every individual entity, the software generates a unique physical profile in memory.

This profile defines the precise geometric boundaries, visual characteristics, and complex collision surface parameters of the object. Crucially, the script assigns high-friction and high-stiffness coefficients to the outer collision mesh of each sphere, accurately mimicking the rough, baked-clay exterior of an actual seedball.

Once the mathematical profile is compiled into a valid simulation description format, the software issues a subsystem command to inject the new entity directly into the active physics environment for immediate collision processing.

7.2.2 Controlled Spatial Distribution

To prevent the physics engine from failing due to instantaneous mesh overlap—which causes violent, infinite-force expulsions in rigid-body solvers—the software employs a three-dimensional spatial offsetting algorithm.

The script calculates precise Cartesian coordinates to organize the incoming payload into a structured grid pattern suspended above the hopper.

7.3 Dynamic Physics Assignment and Flow Diagnostics

The core intelligence of the testing script lies in its mathematical distribution algorithms. These algorithms ensure the simulation stresses the hopper design exactly as a diverse biological payload would in a real-world reforestation mission.

7.3.1 Variance Modeling for Jam Prediction

Every generated entity undergoes a randomized property assignment based on predefined biological variances ranging from 15 mm to 35 mm. The software utilizes uniform randomization algorithms to dictate the specific diameter d of each object:

$$d \sim \mathcal{U}(15 \text{ mm}, 35 \text{ mm})$$

Once the randomized geometric size is established, the script automatically derives the exact physical properties required by the rigid-body solver.

Volume and Mass The volume V of each spherical seedball is calculated as:

$$V = \frac{4}{3}\pi r^3$$

The mass m is then derived using a baseline biological density:

$$\rho = 1500 \text{ kg/m}^3$$

$$m = \rho V$$

Moment of Inertia The rotational inertia matrix I for a solid sphere is:

$$I = \frac{2}{5}mr^2$$

This dictates how uniquely the object tumbles and resists angular momentum changes as it falls.

By embedding these unique physics properties into each object, the software creates a highly realistic, chaotic flow scenario. Differently weighted and sized objects interact, interlock, and collide at varying velocities, which is the primary cause of granular bridging, arching, and mechanical jamming at the hopper's throat.

7.3.2 Visual Analytics Mapping

To assist the engineering team in analyzing sorting behavior and identifying exact jam locations, the software includes a visual mapping algorithm. It mathematically normalizes the randomized size data and translates it into an RGB color gradient:

- **Small Entities:** Assigned one end of the color spectrum (e.g., green).
- **Large Entities:** Assigned the contrasting end (e.g., red).
- **Median Entities:** Assigned transitional gradient colors (e.g., yellow/orange).

This creates a clear, immediate visual indicator of mass distribution throughout the payload. By observing the color gradient during the simulation, operators can instantly verify whether the hopper successfully funnels all sizes evenly. More importantly, it visually demonstrates whether large masses (red spheres) preferentially lock together at the dispensing bottleneck to cause a jam, or if the system suffers from granular convection—where smaller seedballs (green spheres) improperly filter to the bottom while larger ones float to the top.

8 Analysis

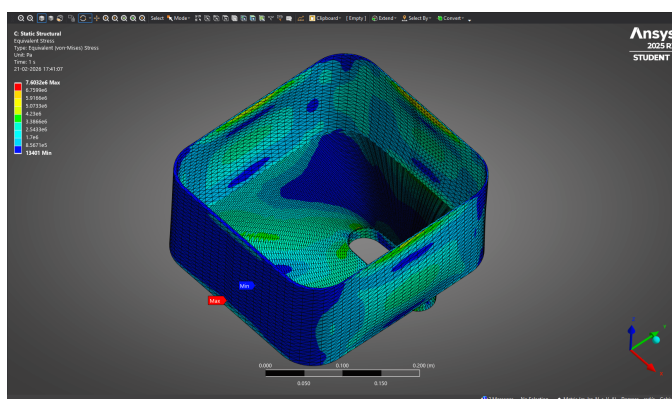
8.1 FEM & FEA

Finite Element method is a numerical technique used to find accurate solutions to boundary value problems for partial differential equations. Basically, a complex geometry is transformed into a mesh composed of several small shapes called elements (most commonly tetrahedrons or triangles). These elements are connected at specific points called nodes. FEM converts the PDEs (which are extremely difficult to solve, given any geometry) into a system of algebraic equations that a computer can solve with relative ease.

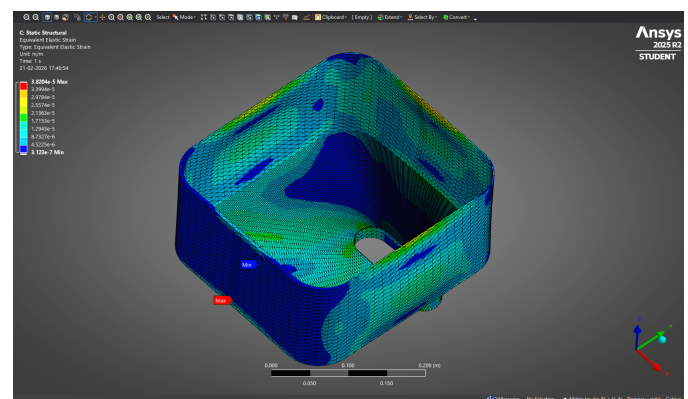
Finite Element Analysis is the practical application of FEM to simulate how a design reacts to real-world forces. In the context of our design, FEA is used to predict how the weight of material (static load) affects the structural integrity. FEA predicts if the structure will bend or break under its own static load. It allows for iterative design meaning you can thin out a part to save weight without ever touching a physical prototype.

8.1.1 Equivalent Stress Analysis:

Equivalent stress analysis essentially decomposes a 3D stress state into a single scalar value and maximum principal stress finds the maximum normal stress acting on a plane where the shear stress is zero. The values show that there is no pile-up of forces under the weight of the structure itself, ensuring that the structure is stable and is not prone to suffering failure due to stress at any point on the design. The values obtained for strain also show that there is no significant deviation from the structure's original shape and thus we can say that the strain on the design is also very well controlled.



(a)

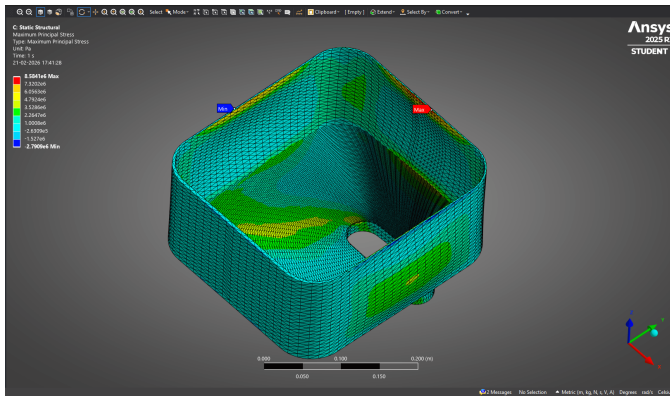


(b)

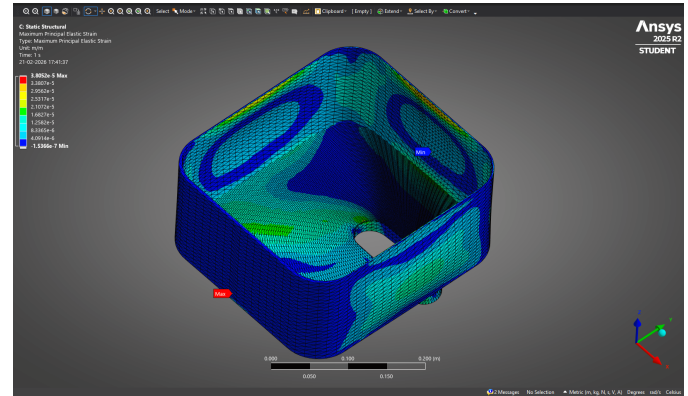
Figure 8: Equivalent Stress-Strain

8.1.2 Maximum Principal Stress-Strain

For maximum principal stress, the values and the contour map also indicate that even under the worst-case scenario load, our design is not at risk of failure. Also to note, is that the maxima of the values calculated under FEA show that the region suffering the most stress is not where the seedballs will regularly interact or strike. For the maximum principal strain analysis, it should be noted that even under the most dire conditions, our design would hold up.



(a)

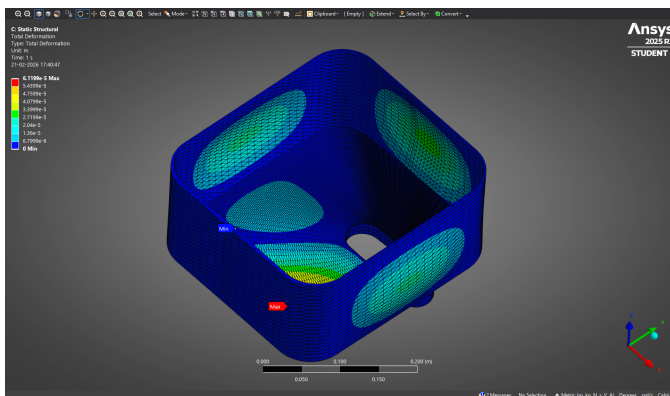


(b)

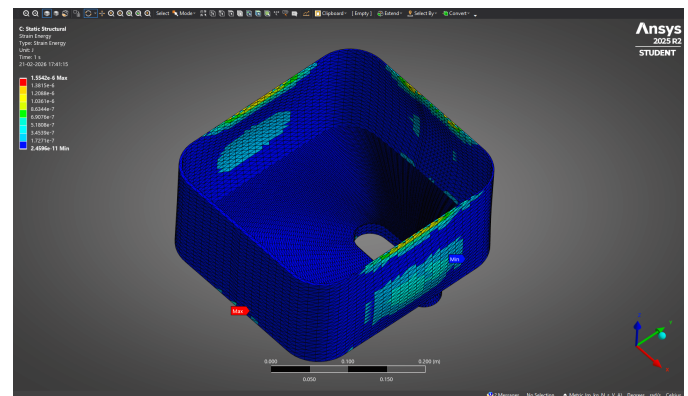
Figure 9: Maximum Principal Stress-Strain

8.1.3 Volume Deformation:

Volume deformation refers to the change in physical space an object occupies when subjected to stress. Strain energy is the work done by external loads to deform a material within its elastic limit, which is recoverable upon unloading. The contours show that although the hopper suffers from deformation, it is not significant and the values clearly show that no failure could possibly occur.



(a) Volume Deformation



(b) Strain Energy

Figure 10: Volume Deformation and Strain Energy

8.1.4 Strain Energy

Following our discussion deformation, the energy that is subsequently stored in the structure is shown as above. Even at its maxima, the values are not significant and show no risk of catastrophic failure.

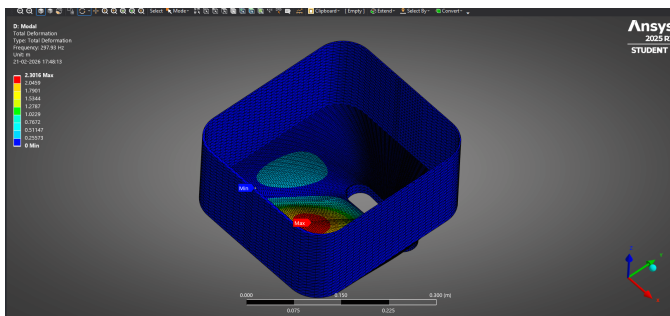
8.2 Modal Analysis

Modal Analysis Explanation:

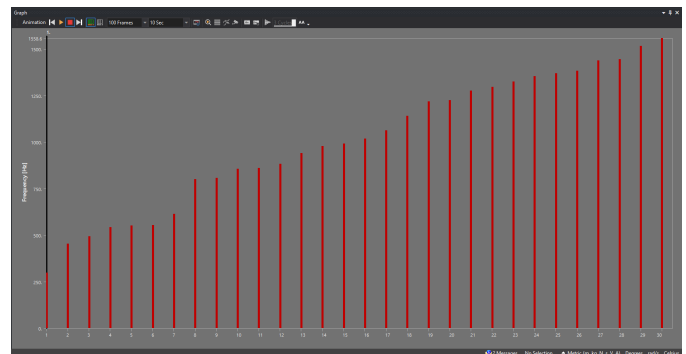
Modal analysis is a linear dynamic analysis used to determine the natural frequencies and mode shapes of a structure. It is critical for avoiding high-amplitude vibrations inside the structure and suffering subsequent chances of failure. Mode shapes predict the physical shape the structure will take at some frequency. This analysis allows for avoiding catastrophic failure where high amplitude vibrations cause any of the components used to break, buckle or shear.

Modal Analysis Simulation:

The key reason for modal analysis for any system is to avoid high amplitude vibrations occurring due to resonant frequencies that are produced by both internal and external factors. The given graph, contour map and values lay out a complete modal analysis of the design. Persistent vibrations of the given frequency are of course, avoided as far as possible (stemming from internal components such as the servos). However, even if momentary high amplitude vibrations occur, the design does not go through catastrophic failure and lies well within safe limits.



(a) Deformation due to Frequencies



(b) Frequency Analysis

Figure 11: Maximum Principal Stress-Strain

It is to be noted that the design is constructed with a safety factor of 2. The avoidance of failure is dependent on this, and it is not always mentioned explicitly.

9 Jamming

The reliability of the seedball dispensing mechanism is primarily dictated by its ability to maintain continuous flow despite the stochastic nature of the payload. As highlighted in the Problem Statement, the seedballs are non-uniform, irregular spheres ranging from 15 mm to 35 mm in diameter. This high variability, combined with the significant overburden pressure from the 15 kg payload, creates a high probability of flow obstruction. Jamming in this system is not a single phenomenon but rather a result of two distinct failure modes: static bridging at the hopper throat and dynamic mechanical interlocking within the metering mechanism.

9.1 Geometric Interlocking

Geometric interlocking, often referred to as "arching" or "bridging," occurs at the transition zone where the bulk storage (hopper) narrows into the metering throat (the obround outlet).

1. **Mechanism of Failure:** Due to the irregular surface texture and shape of the seedballs. When multiple large seedballs (approx. 30-35 mm) arrive at the discharge outlet simultaneously, they can mechanically lock against each other and the hopper walls to form a stable structural arch.
2. **The "Keystone" Effect:** Once these arch forms, it supports the entire weight of the seed column above it, effectively shielding the dispensing roller from the payload.

9.2 Sequential Jamming

Sequential jamming is a dynamic failure mode specific to the two parallel V-file dispensing channels. Unlike the static arching at the hopper throat, this jamming occurs along the length of the dispensing tracks.

1. **Mechanism of Failure (Wedging):** The V-profile is designed to align seeds in a single file. However, due to the high variance in seedball diameter (15–35 mm) and irregular sphericity, a large or oddly shaped seed can become stuck between the inclined walls of the V-channel. This interference fail creates a friction lock that exceeds the gravitational or vibrational force intended to move the seed.
2. **The "Stacking" Effect:** Once a seed gets stuck in one of the two files, it acts as a physical barrier and once a seed is lodged in the channel, other seeds fall down due to gravity but cannot move past the lodged one. As they accumulate behind the lodged one, their kinetic energy is transferred to the stationary seed, thus increasing the compressive force on it. Eventually, a free-flowing stream of seeds transforms into a compacted and interlocked pile of seeds. In the end, the channel changes from a dynamic flow to a stationary, high-friction plug, thus increasing the chances of a jam.
3. **Multi-Channel Dependency:** Since the two V-files are supplied by a common hopper outlet, a blockage in one channel directly affects the overall flow behaviour of the system. When one path becomes obstructed, a greater portion of the incoming seedballs is redirected into the remaining open channel. This uneven feeding can increase particle crowding and interaction within the active path, raising the likelihood of further obstruction and cascading jamming events. Further the jamming effect of arching can also take place here also.

10 Mitigation

To keep the irregular seedballs flowing smoothly, the system needs two things: mechanical movement to break blockages and proper hopper design to prevent locking.

Active Vibration: - To prevent the "Keystone Effect" explained in Section 8.1, the hopper outlet needs an active system to stop stable arches from forming.

Vibratory Excitation: - Using a low-frequency vibrator helps prevent both static bridging and repeated jamming inside the hopper and dispensing channel. High-frequency vibration can damage the structure or press the material tighter. Low-frequency vibration creates larger movements that break arches and friction locks without putting too much stress on the structure.

Installation Of V-file: - This design change divides the hopper outlet into two V-shaped channels. This improves flow control and reduces interference between seedballs. By splitting the payload into two smaller streams, the load and pressure in each channel decrease. This reduces the chance of seedballs locking together or forming arches. The V-shape also helps guide and align irregular seedballs, reducing random movement and wedging.

Jam Detection and Rectification Logic: - An IR (infrared) sensor-based mass flow detection system is used to monitor seed discharge continuously. The sensor is placed near the outlet to detect seedball movement. During normal operation, the sensor detects regular interruptions as seedballs pass through. If the flow rate drops a lot or no movement is detected for a fixed time, the system treats this as a jam or bridging problem at the hopper outlet.

Geometric and Material Optimization: - For a straight-line hopper, the following changes should be made to ensure smooth mass flow instead of funnel flow or bridging: To prevent the Keystone Effect, the hopper outlet width (W) should be at least 2.5 to 3 times the diameter of the largest seedball (d-max). Since d-max is 35 mm, the outlet width should be at least 90–105 mm. The hopper walls should be inclined at an angle greater than 60° from the horizontal. This ensures that gravity is strong enough to overcome friction between the seedballs and the walls.

Redundant Locking and Leakage Prevention: - The dispensing system should stay closed by default if power is lost or a sensor fails. Spring-loaded flaps or high-friction rotary designs ensure that gravity alone cannot release the payload by accident.

Communication and Pilot Notification: - If a fail-safe condition is activated, a high-priority signal is sent to the flight controller or ground station. This allows the operator to stop the mission and safely recover the UAV before serious mechanical or electrical damage happens.

11 Risk Analysis And Failure Modes (FMEA)

11.1 Identification of Potential Failure Modes

Servo Motor Stall and Gear Failure: - The servo motor is used to open and close the flap to control the release of the payload. If the flap gets blocked or cannot move freely, the servo motor will still try to move to its set position. If there is no sensor or protection system to detect this problem, the motor will continue running under load. This can cause the motor to stall, damage the internal gears, or overheat. If the servo fails completely, the flap may suddenly open or stop working properly. This can lead to an uncontrolled release of the full 15 kg payload and may disturb the balance of the system.

Vibration-Induced Structural Fatigue: - During operation, continuous vibration occurs, especially when carrying a 15 kg payload. These vibrations create repeated stress on the frame and fasteners. Over time, this repeated stress can weaken the material. If the vibration frequency becomes close to the natural frequency of the structure, resonance can happen. In resonance, even small vibrations can cause large movements. This can increase stress on the frame, brackets, and joints. Over long use, screws can loosen, threaded connections can weaken, and important joints may lose tightness.

11.2 Risk Assessment

Assessment of Hopper Flow Risks: -

1. **Likelihood:** Flow blockage in a straight-line hopper is high because the design does not fully support smooth material flow.

Straight-line hoppers often cause funnel flow instead of full material flow. This creates dead zones near the walls. As the payload becomes less during operation, the internal pressure changes and reduces the force needed to keep the material moving.

2. **Impact:** Blockage or bridging inside the hopper is serious because it stops the system from working properly. When material cannot come out, the whole deployment process stops. This wastes battery power, uses extra flight time, and does not fully use the payload.

Assessment of Servo Actuation Risks: -

1. **Likelihood:** Medium–High. The wide size distribution of seedballs significantly increases the chance of obstruction during flap closure over extended missions.
2. **Impact:** Catastrophic. Permanent servo failure can cause uncontrolled flap opening, resulting in an instantaneous 15 kg payload dump that destabilizes the UAV and may lead to a crash.

Assessment of Vibrator and Structural Risks: -

1. **Likelihood:** Continuous vibration can slowly loosen screws and mounting parts over time. If the vibration frequency matches the natural frequency of the structure, resonance can increase vibration and stress on the frame and joints.
2. **Impact:** Loose structure may not cause immediate failure but can reduce performance and reliability. Loose brackets or mounts may cause small misalignment, leakage, or uneven movement.

Assessment of Payload Characteristics Risk: -

1. **Likelihood:** High. Irregular seedball geometry produces unpredictable interactions with the aluminium dispensing flaps.
2. **Impact:** Moderate. Inconsistent drop rates and uneven hinge loading accelerate mechanical wear, increasing maintenance frequency.

12 Future Scope

Although the current straight-line hopper with dual V-file mechanism works properly, there is still good scope to improve the system. In the future, the design can be made smarter, lighter, more reliable, and more automatic so that it can handle large-scale missions more efficiently. The following improvements can be considered:

1. Smart Flow Control System

Adaptive Drop Timing: In future versions, the system can automatically adjust the servo opening and closing time based on how fast the seeds are moving inside the V-file and how fast the drone is flying. This will help maintain equal spacing between seedballs on the ground. Even if the drone speed changes due to wind or pilot control, the system will still maintain uniform distribution.

Speed-Based Automatic Control: The dispenser can be connected directly to the drone's GPS and speed data. When the drone moves faster, the drop rate can increase automatically. When it slows down, the drop rate can decrease. This will remove the need for constant manual adjustment by the pilot and improve planting accuracy.

Improved Flow Monitoring: More advanced sensors can be installed to continuously monitor seed movement from the hopper to the gate. This will help in detecting irregular flow conditions early and improving overall drop precision.

2. Advanced Jam Detection and Prevention

Early Jam Prediction System: Instead of waiting for a complete blockage, the controller can study small changes in sensor signals such as delay in seed movement. If it detects unusual patterns, it can take action before a full jam occurs. This will reduce flight time loss and improve mission success rate.

Automatic Self-Clearing Action: The servo gate and V-file can be programmed to perform small periodic shaking movements during operation. These small movements will prevent seed buildup, and lower the chance of seeds getting stuck inside the channel.

Backup Sensor System: Additional IR or small proximity sensors can be placed near the hopper outlet and at the end of the V-file. If one sensor becomes defective or stops working due to dust or mud, the second sensor can continue monitoring the flow. This increases system safety and reliability during real field operations.

3. Hopper Geometry Improvement

Computer-Based Flow Testing: Special software can be used to simulate how seedballs move inside the hopper. This allows engineers to test different wall angles and slot sizes on a computer before building the actual model. It saves time, cost, and improves design accuracy.

Adjustable Slot Mechanism: The slot opening can be made adjustable so that the same system can handle different seedball sizes. This will make the dispenser more flexible for different types of seeds.

Low-Friction Surface Treatment: Applying smooth coatings inside the hopper can reduce sticking, especially when seedballs contain moist soil or clay. This will improve flow consistency and reduce cleaning requirements.

4. Material and Structural Enhancement

Lightweight Strong Materials: In the future, aluminium parts can be replaced with lighter materials such as carbon fiber or reinforced plastic. This will reduce the total weight of the drone and increase flight time.

Better Vibration Control: Improved mounting systems with damping materials can reduce vibration transfer from the dispenser to the drone frame. This will protect electronic components and improve flight stability.

Modular and Easy Maintenance Design: The hopper and V-file assembly can be designed as a detachable unit. This will make cleaning, inspection, and replacement faster and easier during field missions.

5. Autonomous and GPS-Based Seeding

Location-Based Drop Control: By using GPS mapping, the drone can automatically change the seed drop

density depending on land type or preloaded data.

Terrain-Based Automatic Adjustment: Cameras or onboard sensors can analyze the land below. The system can then adjust drop rate for forest gaps, farmland, or open land without pilot intervention.

Fully Automatic Operation: The entire mission, including flying and seed dropping, can be automated using flight planning software. This will reduce pilot workload and improve accuracy in large reforestation projects.

6. Scalability and Multi-Mission Adaptability

Multiple Hopper System: Future designs can include more than one hopper working together. This will increase total seed capacity and allow different seed types to be deployed in a single mission.

Cartridge-Based Refilling: Removable seed cartridges can be developed so that refilling becomes quick and simple. This will reduce turnaround time between missions.

Large-Scale Industrial Version: The design can be expanded for bigger drones that carry heavier payloads. This will support large reforestation and agricultural projects.

7. Energy and Power Management

Separate Power System for Electronics: A dedicated power supply for servos and sensors can prevent voltage drops and unexpected system resets during operation.

Smart Battery Monitoring: The system can continuously monitor battery level and adjust drop rate if power becomes low. This ensures safe mission completion and safe return of the drone.

In the future, these improvements can make the seed dispensing system more intelligent, reliable, lightweight, and suitable for large-scale agriculture, reforestation, ecological restoration, and disaster recovery missions.

13 Conclusion

In aerial seed dropping, the main challenge is to release only one seedball at a time. Since the seedballs are not of the same size and shape, they may fall together, get stuck, or drop more than one at once. This affects proper spacing and reduces accuracy.

The developed straight-line hopper with dual V-files solves this issue effectively. The hopper is designed with steep wall angles and smooth inner surfaces so that all the seedballs keep moving downward without forming dead zones. A rectangular slot outlet is used instead of a circular hole, which reduces the pressure at the bottom and lowers the chance of arch formation. This slot spreads the seedballs in a line before they enter the V-files, reducing compression and friction between them. The two V-files then guide the seedballs into a single straight path so they do not move side by side. At the end of each V-file, a servo-controlled gate is installed which stays normally closed for safety. An IR sensor checks if a seedball is ready before the gate opens. The servo then performs a quick open–close action to release only one seedball at a time. Overall, the integration of optimized hopper geometry, dual V-file alignment, intelligent sensing, and a precise servo-based gate system results in a compact, lightweight, and reliable dispensing mechanism. The design successfully balances mechanical simplicity, flow physics, and electronic control, making it suitable for safe and efficient aerial reforestation missions.

References

- [1] Darias, J.R., Gella, D., Fernández, M.E., Zuriguel, I., and Maza, D., 2020. "The hopper angle role on the velocity and solid-fraction profiles at the outlet of silos." *Powder Technology*, vol. 366, pp. 488–496.
- [2] Jenike, A.W., and Johanson, J.R., 1969. "On the Theory of Bin Loads." *Journal of Engineering for Industry*, vol. 91, no. 2, pp. 339–344.
- [3] Kidanemariam, A.G., Chan-Braun, C., Doychev, T., and Uhlmann, M., 2013. "Direct numerical simulation of horizontal open channel flow with finite-size, heavy particles at low solid volume fraction." *New Journal of Physics*, vol. 15, no. 2, 025031.
- [4] Bembenek, M., Dmytriv, V., Banha, V., Horodniak, R., and Pawlik, J., 2024. "A Mathematical Model for Conical Hopper Mass Efficiency." *Applied Sciences*, vol. 14, no. 16, 7373.
- [5] Balevičius, R., Kačianauskas, R., Mróz, Z., and Sielamowicz, I., 2011. "Analysis and DEM simulation of granular material flow patterns in hopper models of different shapes." *Advanced Powder Technology*, vol. 22, no. 2, pp. 226–235.
- [6] Huang, X., Zheng, Q., Liu, D., Yu, A., and Yan, W., 2022. "A design method of hopper shape optimization with improved mass flow pattern and reduced particle segregation." *Chemical Engineering Science*, vol. 253, 117579.
- [7] Shinohara, K., Idemitsu, Y., Gotoh, K., and Tanaka, T., 1968. "Mechanism of gravity flow of particles from a hopper." *Industrial & Engineering Chemistry Fundamentals*, vol. 7, no. 3, pp. 383–387.